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Title : Traffic Light Controller

GROUP - B-12

Name - Khondoker Khalid Hasan Kaif

Roll - 2309056

Batch - 2k23

&

Name - Tasfia Tohura

Roll - 2309057

Batch- 2k23

Abstract

This project presents the design and simulation of a digital traffic light management system using standard TTL integrated circuits. The system employs a 555 timer IC as an astable oscillator to generate clock pulses, which drive a 7490 decade counter configured to count from 0 to 6. A 7411 three-input AND gate detects the terminal count and resets the counter, while a 7447 BCD-to-seven-segment decoder drives a common-anode seven-segment display to provide a visual countdown. A separate 555 timer circuit, combined with two D-type flip-flops (DTFF) and an AND gate, controls the traffic light LEDs following the standard United Kingdom traffic signal sequence: Red, Red and Yellow, Green, Yellow, then back to Red. The project demonstrates practical application of sequential and combinational logic, modular circuit design, and real-world timing considerations using widely available, low-cost components.

1. Objectives

The primary objectives of this project are as follows:

1. To design and implement a functional traffic light control circuit using standard digital TTL ICs, including the 555 timer, 7490 counter, 7411 AND gate, and 7447 decoder.
2. To generate a stable clock signal using the 555 timer configured in astable mode and use it to drive a sequential counting circuit.
3. To implement a BCD counter that counts from 0 to 6 and automatically resets upon reaching 7, providing a repeating 7-state timing cycle.
4. To decode the BCD count output and display the corresponding digit (0–6) on a seven-segment display, providing a visual countdown for pedestrians or drivers.
5. To implement the UK standard traffic light sequence (Red → Red+Yellow → Green → Yellow → Red) using D-type flip-flops and combinational logic.
6. To demonstrate the modular design approach by separating the timing/counting section from the traffic light control section.
7. To evaluate the performance of the system, identify its limitations, and suggest areas for future improvement.

2. Introduction

Traffic management systems are a fundamental component of modern urban infrastructure. At their core, these systems rely on precisely timed sequences of light signals to regulate the flow of vehicles and pedestrians at road junctions. While contemporary traffic management increasingly depends on microcontrollers and software-driven systems, understanding the underlying digital logic principles remains an essential foundation for electronics and computer engineering students.

This project explores the design of a traffic light control system built entirely from discrete digital integrated circuits (ICs), without the use of any microcontroller or programmable logic device. The design leverages the 555 timer IC — one of the most versatile and widely used ICs in electronics — alongside standard TTL logic ICs including the 7490 decade counter, the 7447 BCD-to-seven-segment decoder, the 7411 three-input AND gate, and D-type flip-flops.

The system operates in two parallel sections: a counting and display section, which drives a seven-segment display showing a digit count from 0 to 6, and a traffic light control section, which sequences the red, yellow, and green LEDs according to the UK traffic signal convention. Both sections are driven by independently configured 555 timer oscillators, demonstrating a modular design approach.

The project was designed and simulated using Proteus, a professional electronics design automation (EDA) tool. The schematic diagram of the complete circuit is presented in Section 4 of this report.

3. Background Theory

3.1 The 555 Timer IC (Astable Mode)

The NE555 (or simply '555') timer IC is a versatile integrated circuit capable of producing stable time delays or oscillations. When configured in astable mode, it operates as a free-running oscillator, continuously producing a square wave output without requiring any external trigger. The frequency and duty cycle of the output are determined by two external resistors (R_a , R_b) and a timing capacitor (C). The charge time is $t_1 = 0.693 \times (R_a + R_b) \times C$, the discharge time is $t_2 = 0.693 \times R_b \times C$, giving a total period $T = 0.693 \times (R_a + 2R_b) \times C$ and frequency $f = 1.44 / ((R_a + 2R_b) \times C)$.

In this project, two independent 555 timers are used. For the counting/display circuit (U5): $R_a = R_3 = 100 \text{ k}\Omega$, $R_b = R_4 = 22 \text{ k}\Omega$, $C = C_2 = 10 \text{ }\mu\text{F}$, giving $T \approx 0.997 \text{ s}$ and $f \approx 1.00 \text{ Hz}$. For the traffic light circuit (U4): $R_a = R_1 = 10 \text{ k}\Omega$, $R_b = R_2 = 100 \text{ k}\Omega$, $C = C_1 = 47 \text{ }\mu\text{F}$, giving $T \approx 6.84 \text{ s}$ and $f \approx 0.146 \text{ Hz}$. Each traffic light phase therefore lasts approximately 6.84 seconds before advancing to the next state.

3.2 The 7490 Decade Counter

The 7490 is a TTL MSI decade counter capable of counting from 0 to 9 in BCD (Binary Coded Decimal) format across its four output pins Q0–Q3. It features two clock inputs (CKA and CKB), which must be cascaded for standard BCD counting, and asynchronous reset inputs R0(1), R0(2) (reset to 0) and R9(1), R9(2) (reset to 9). In this design, the counter is configured to count from 0 to 6. The output Q1 and Q2 are fed into the 7411 AND gate; when the count reaches 7 (binary 0111), the AND gate output goes high, asserting the reset line and returning the counter to 0. This creates a modulo-7 counter (states 0 through 6).

3.3 The 7447 BCD-to-Seven-Segment Decoder

The 7447 IC is a BCD-to-seven-segment decoder/driver designed specifically to drive common-anode seven-segment displays. It takes a 4-bit BCD input (A, B, C, D) and produces active-low outputs (a–g) that directly control the seven segments of the display. The 7447 also features a Blanking Input/Ripple Blanking Output (BI/RBO) and a Ripple Blanking Input (RBI) for multi-digit display blanking. In this project, the four BCD outputs (Q0–Q3) of the 7490 counter are connected directly to the A–D inputs of the 7447, and its outputs drive the seven-segment display to show digits 0 through 6.

3.4 D-Type Flip-Flops

A D-type flip-flop (DTFF) is a fundamental sequential logic element that captures the value of its data input (D) on the rising edge of a clock signal and holds that value at its output (Q) until the next clock edge. Two D flip-flops (U6 and U8) are used in the traffic light section of this project to implement a simple 2-bit state machine. The combination of their Q outputs, processed through an AND gate (U7), produces the correct logic signals to drive the three traffic light LEDs (red, yellow, green) in the appropriate UK sequence.

3.5 UK Traffic Light Sequence

The United Kingdom standard traffic light sequence consists of four distinct states, cycling as follows:

- Red — vehicles must stop
- Red + Yellow — vehicles prepare to move
- Green — vehicles may proceed
- Yellow — vehicles prepare to stop

This four-phase sequence is implemented using the 2-bit state output of the flip-flops, where each combination of Q outputs maps to one of the four traffic light states.

4. Circuit Design and Schematic

4.1 Circuit Schematic

Figure 1 below shows the complete Proteus simulation schematic of the Traffic Light Management System, including both the counting/display section (upper half) and the traffic light control section (lower half).

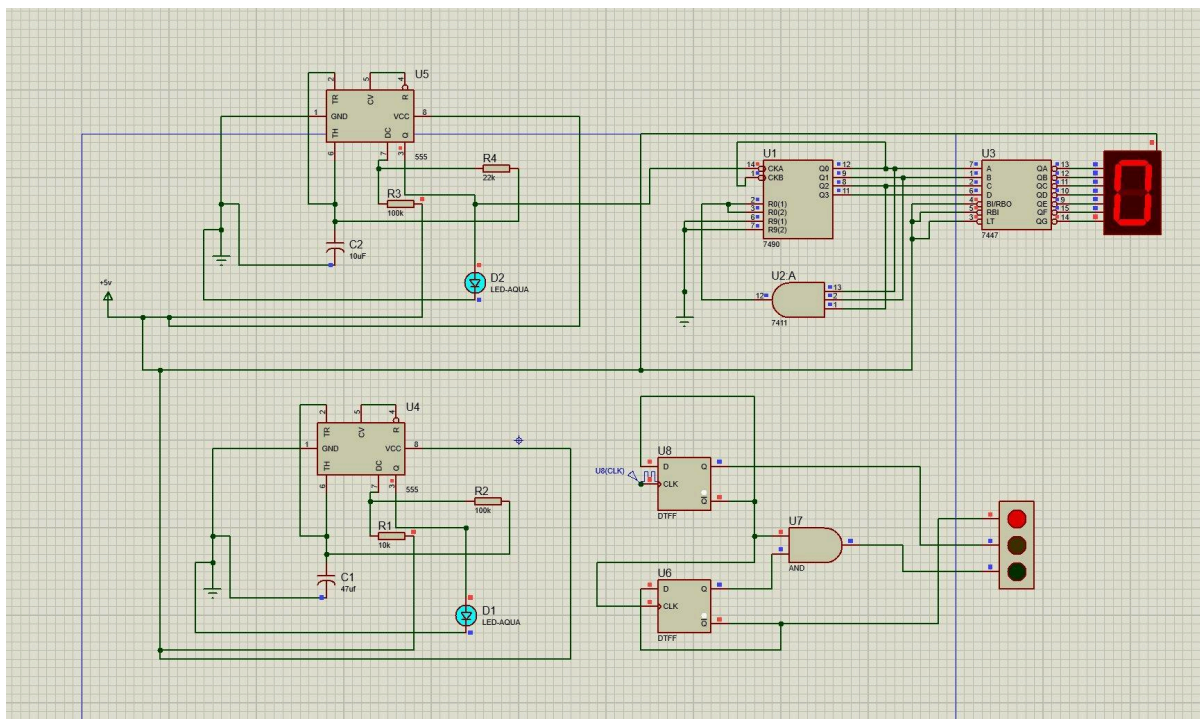


Figure 1: Proteus simulation schematic of the Traffic Light Management System

4.2 Component List

Ref.	Component	Function in Circuit
U4, U5	NE555 Timer	Astable oscillators generating clock pulses for each subsystem
U1	7490 Decade Counter	Counts 0–6 in BCD; resets on reaching 7 via AND gate feedback
U2:A	7411 AND Gate (3-input)	Detects count = 7 ($Q1 \cdot Q2 \cdot Q3 = 1$) to reset the 7490 counter
U3	7447 Decoder/Driver	Converts BCD output of 7490 into 7-segment display signals
U6, U8	D-Type Flip-Flop (DTFF)	Implements 2-bit state machine for traffic light sequencing
U7	AND Gate	Combines flip-flop outputs to produce correct traffic light signals
R1	10 k Ω Resistor	Timing resistor for 555 (U4) — traffic light oscillator
R2	100 k Ω Resistor	Timing resistor for 555 (U4) — traffic light oscillator
R3	100 k Ω Resistor	Timing resistor for 555 (U5) — counter oscillator
R4	22 k Ω Resistor	Timing resistor for 555 (U5) — counter oscillator
C1	47 μ F Capacitor	Timing capacitor for 555 (U4)
C2	10 μ F Capacitor	Timing capacitor for 555 (U5)
D1, D2	LED (Aqua)	Visual indicators of 555 oscillator output
—	7-Segment Display	Displays count value 0–6 driven by 7447 decoder
—	Traffic Light LEDs	Red, yellow, and green LEDs representing traffic signals

Table 1: Component list and functions

5. System Operation

5.1 Counting and Display Subsystem

The upper section of the circuit operates as follows: the 555 timer (U5), configured in astable mode with $R3 = 100\text{k}\Omega$, $R4 = 22\text{k}\Omega$, and $C2 = 10\mu\text{F}$, produces a periodic clock signal. This clock drives the CKA input of the 7490 decade counter (U1), with CKB connected to Q0 to enable standard BCD counting. The counter increments its output (Q0–Q3) on each falling edge of the clock.

The Q1 and Q2 outputs of the 7490 are fed into the three-input AND gate (U2:A, 7411). When the count reaches 7 (binary 0111 — where $Q0=1$, $Q1=1$, $Q2=1$), the AND gate output goes high, driving the R0(1) and R0(2) reset pins of the 7490. This causes the counter to immediately reset to 0, creating a modulo-7 counter that continuously cycles through the states 0, 1, 2, 3, 4, 5, 6, and back to 0.

The BCD outputs Q0–Q3 are simultaneously connected to the A–D inputs of the 7447 BCD-to-seven-segment decoder (U3). The 7447 decodes each BCD value and drives the common-anode seven-segment display, showing the digits 0 through 6 in synchrony with the counter cycle.

5.2 Traffic Light Control Subsystem

The lower section uses a second 555 timer (U4), configured with $R1 = 10k\Omega$, $R2 = 100k\Omega$, and $C1 = 47\mu F$, producing clock pulses at a different rate. These pulses are fed to the clock inputs of two D-type flip-flops (U8 and U6), which together form a 2-bit synchronous state machine.

The Q outputs of U8 and U6 are combined through the AND gate (U7), whose output — together with the individual Q outputs — drives the three traffic light LEDs. The state transitions of the two flip-flops produce the four UK traffic light phases in sequence:

State	Q(U8)	Q(U6)	Lights Active	UK Phase
1	0	0	Red	Stop
2	0	1	Red + Yellow	Prepare to Move
3	1	0	Green	Go
4	1	1	Yellow	Prepare to Stop

Table 2: Traffic light state table

Each state persists for the duration of one full clock cycle of U4, giving each traffic light phase an equal time interval of approximately 7 seconds across the full sequence.

6. Discussion

6.1 Simulation Results

The circuit was simulated in Proteus and performed as designed. The 555 timer (U5) in the upper section produced a stable clock, enabling the 7490 counter to cycle correctly from 0 to 6 before resetting. The seven-segment display accurately showed digits 0 through 6 in sequence, driven by the 7447 decoder. The aqua LED (D2) visually confirmed the oscillation of the upper 555 timer.

In the lower section, the traffic light LEDs sequenced through the UK standard phases (Red → Red+Yellow → Green → Yellow) in a repeating loop. The aqua LED (D1) confirmed the independent oscillation of the traffic light 555 timer (U4). The AND gate (U7) correctly combined the flip-flop outputs to produce the required LED drive signals for each phase.

6.2 Performance Characteristics

The system demonstrated the following operational characteristics:

- Stable, continuous oscillation from both 555 timer circuits with no observed timing drift during simulation.
- Accurate modulo-7 counting: the 7490 counter reset reliably at count 7 without counting beyond 6.
- Correct BCD decoding: the 7447 produced the correct segment patterns for all seven digits (0–6).

- Proper UK traffic sequence: all four phases (Red, Red+Yellow, Green, Yellow) were observed in the correct order.
- Modular independence: the two subsystems operated simultaneously without interfering with each other.

7. Limitations

Despite its successful simulation, the system has four notable limitations:

1. Independent Timing — Cannot Be Adjusted Simultaneously

The counting/display section and the traffic light control section each use their own independently configured 555 timer. Consequently, the two timing circuits cannot be synchronised or adjusted simultaneously using a single control. Changing the timing of one section requires component changes in its respective RC network only, and there is no way to link or lock the two clocks together without additional circuitry.

2. Fixed 7-Second Interval for All States

All four traffic light states share the same clock period from U4, meaning each phase (Red, Red+Yellow, Green, Yellow) is allocated exactly the same duration — approximately 7 seconds per state for the full 4-state cycle. In a real-world traffic system, the green phase typically needs to be significantly longer than the yellow or red+yellow transition phases. This system cannot differentiate phase durations without a substantially more complex timing circuit.

3. Fixed UK Sequence — No Flexibility for Other Configurations

The traffic light sequence is hardwired into the D flip-flop state machine and AND gate logic, implementing only the UK four-phase sequence. The system cannot be easily reconfigured for other international traffic light conventions (such as the US two-phase Red/Green system, or systems with pedestrian phases) without redesigning the logic network. A microcontroller-based or FPGA-based implementation would provide the flexibility to accommodate different sequences through software or programmable logic.

4. Counter Counts Upward Only — No Countdown Capability

The 7490 decade counter is a purely up-counting device, meaning the seven-segment display counts upward from 0 to 6 before resetting. Ideally, for a traffic light timer, the display should count downward from 6 to 0, giving drivers and pedestrians a visual indication of how much time remains in the current phase. Implementing a countdown requires either a presettable down-counter IC (such as the 74190 or 74193) with additional preset logic, or a look-up table approach using combinational logic to reverse the displayed digit — neither of which is straightforward with the 7490/7447 combination used here. This remains a notable functional gap between the prototype and a real-world traffic countdown display.

8. Conclusion

This project successfully demonstrates the design and simulation of a digital traffic light management system using standard TTL ICs. The system integrates two independent 555 timer-based oscillators, a modulo-7 BCD

counter (7490), a BCD-to-seven-segment decoder (7447), a 7411 AND gate for counter reset, and a D flip-flop state machine to control traffic light LEDs through the UK four-phase sequence.

The design highlights the practical application of fundamental digital electronics concepts — including astable oscillators, synchronous sequential logic, BCD counting, and combinational decoding — using simple, low-cost, and widely available components. The modular architecture cleanly separates the counting/display function from the traffic signal control function, making the circuit easier to understand, build, and troubleshoot.

The three identified limitations — inability to simultaneously adjust both timing circuits, fixed equal-duration phases, and a hardwired UK-only sequence — represent clear directions for future improvement. These could be addressed by replacing the dual 555 timers and flip-flop state machine with a programmable microcontroller (such as an ATmega328 or STM32), which would allow dynamic phase timing, multiple sequence configurations, and sensor-based adaptive control.

Overall, the project achieves its educational objectives, providing a practical demonstration of how complex real-world systems — such as urban traffic management — can be implemented and understood through the fundamental principles of digital electronics.

9. Github Link

<https://github.com/KhalidHasanKaif/DELC>